

Field and Galois Theory

Appetizer

1.

Give an example of an algebraic extension which has infinite degree.

* 2.

Suppose that $K \subseteq \mathbb{C}$ is a Galois extension of $\mathbb{Q}$, $[K : \mathbb{Q}] = 4$, and $\sqrt{-m} \in K$ for some positive integer m . Show that $\text{Gal}(K/\mathbb{Q}) \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z}$.

* 3.

Let K be a field, and let $f(x) \in K[x]$ be an irreducible polynomial. Suppose that the splitting field L of $f(x)$ is a Galois extension of K (that is, f is separable). Let $\alpha \in L$ be one of the roots of f .

(a) Show that if $\text{Gal}(L/K)$ is an abelian group, then $L = K(\alpha)$.

(b) Is the converse statement true? This is, suppose $L = K(\alpha)$; must $\text{Gal}(L/K)$ be abelian?

Entrée

* 4.

(a) Give an example of a Galois extension $K/\mathbb{Q}$ with Galois group $\mathbb{Z}/4\mathbb{Z}$, and prove that it is such an example.

(b) Let $K = F_q(t)$, and let $L = F_q(t^{1/p})$ where q is a power of p . Prove that there are no automorphisms of L fixing K .

* 5.

Let $\mathbb{C}(x)$ be the field of rational functions with complex coefficients in the variable x . Thus, x is transcendental over $\mathbb{C}$. Put

$$y = x^n + x^{-n} \in \mathbb{C}(x)$$

for some $n > 0$. Prove that the field extension $\mathbb{C}(x)/\mathbb{C}(y)$ is a finite Galois extension and find its degree.

* **6.**

- (a) Let E be the splitting field over $\mathbb{Q}$ of $e(x) = x^3 + 2x^2 + 3x + 4$, which has discriminant -200 . Find $\text{Gal}(E/\mathbb{Q})$.
- (b) Let F be the splitting field over $\mathbb{Q}$ of $f(x) = x^3 + 4x^2 + 7x + 6$, which has discriminant -72 . Find $\text{Gal}(F/\mathbb{Q})$.
- (c) Find $\text{Gal}(EF/\mathbb{Q})$.

* **7.**

Let K be the splitting field of the polynomial $f(x) = x^4 - x^2 - 1$ over $\mathbb{Q}$. Compute $\text{Gal}(K/\mathbb{Q})$. (For partial credit, find the degree $[K : \mathbb{Q}]$.)